

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech. Branch : Mechanical Engineering and Allied/Automation And Robotics/Robotics/Mechatronics Engineering

Semester : IV

Subject Code & Name: BTMPE405A, Numerical Methods in Engineering

Max Marks: 60

Date: 31/12/2024

Duration: 3 Hrs

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q1	Objective type questions. (Compulsory Question)		12
1	In a measurement done by a regular foot scale, the maximum significant digits will be a. ____ 2 ____ b. ____ 3 ____ c. ____ 4 ____ d. ____ 1 ____	Understand/CO1	1
2	The relative error of the number 8.6 if both the digits are correct is a. ____ 0.05 ____ b. ____ 0.01 ____ c. 0.0058 ____ d. 0.5 ____	Understand/CO1	1
3	To apply bisection method for finding out the real root, one of the conditions is a. function should be continuous b. function should be discontinuous c. both continuous and discontinuous d. none of the above	Memory/CO2	1
4	Newton Raphson convergence is a a. Linear process b. quadratic process c. third order d. None of the above	Memory/CO2	1
5	1/3 rd Simpson method is a a. First order accurate b. second order accurate c. third order accurate d. fourth order accurate	Memory/CO2	1
6	Forward elimination in the naïve Gauss elimination method convert the coefficient matrix in		1

	a. Upper triangular matrix	b. Unit matrix	c. Lower triangular matrix	d. None of the above	Memory/CO2	
7	While finding the root of the equation $f(x) = x^3 - 2x - 5$ by using bisection method with lower limit $x_l = 2$ and upper limit $x_u = 3$, the upper limit value in the second iteration would be				Analysis/CO2	1
	a. 2	b. 3	c. 4	d. 2.5		
8	Saturation temperatures are 100 °C and 110 °C at 1 bar and 1.5 bar respectively. The values of saturation temperature at 1.2 bar employing linear interpolating would be				Analysis/CO3	1
	a. 106 °C	b. 104 °C	c. 103 °C	d. 105 °C		
9	The number of segments in the multiple Simpson's 1/3 rule method is in				Memory/CO2	1
	a. even number	b. Odd number	c. may be odd or even number	d. infinite		
10	Rounding off the number 15462 to three significant digits is				Understanding/CO1	1
	a. 15600	b. 15500	c. 15400	d. 15450		
11	The error in the Central difference approximation of the first derivatives is the order of with h as the step size				Analysis/CO2	1
	a. (h)	b. $(h)^2$	c. $(h)^3$	d. none of the above		
12	Linear regression fits a				Memory/CO2	1
	a. Curve	b. Straight line	c. Both	d. Hyperbola		
Q.2 Solve the following.						12
A) Explain the concept of Accuracy and precision with suitable example and illustration and also their interrelation with bias and uncertainty.						Understanding/CO1 6
B) If $u = 3v^7 - 6v$, find the percentage error in u at $v = 1$, if the error in v is 0.05.						Analysis/CO1 6
Q.3 Solve the following.						12
A) Find the real root of the equation employing bisection method $xe^x = 1$ up to $\epsilon_a = 5\%$, which lies between 0 and 1.						Application/CO2 6
B) Obtain the expression of Newton-Raphson method from first principle supported with proper illustration.						Analysis/CO2 6

Q. 4	Solve Any Two of the following.		12								
A)	Explain the concept of scaling, pivoting and singularity wrt to solution of simultaneous algebraic linear equations	Understanding/CO2	6								
B)	Given the system $\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$ Solve by naïve Gauss elimination. Show all steps of the computation.	Analysis/CO2	6								
C)	Given the system $\begin{aligned} x_1 + x_2 - x_3 &= -3 \\ 6x_1 + 2x_2 + 2x_3 &= 2 \\ -3x_1 + 4x_2 + x_3 &= 1 \end{aligned}$ Solve by Cramer Rule. Show all the steps of computation.	Analysis/CO2	6								
Q.5	Solve Any Two of the following.		12								
A)	Obtain the expression of the Numerical Integration Technique Multiple Trapezoidal rule with proper illustration. https://www.batuonline.com	Analysis/CO2	6								
B)	The work done on an object is equal to the force times the distance moved in the direction of the force. The velocity of an object in the direction of a force is given by $v = 4t \quad 0 \leq t \leq 6$ where v is in m/s. Employ the multiple application trapezoidal rule to determine the work if a constant force of 200 N is applied for all t .	Apply/CO3	6								
C)	Use Simpson's 3/8 rule to integrate the function $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from the limits $a = 0$ to $b = 0.8$.	Analysis/CO2	6								
Q. 6	Solve Any Two of the following.		12								
A)	Solve the following ODE using Euler's method from $t = 0$ to 3 taking step size of 0.5 . $\frac{dy}{dt} = -y + t \quad \text{given } y(0) = 1$	Analysis/CO2	6								
B)	The table below gives the temperature T ($^{\circ}\text{C}$) and length l (mm) of a heated rod. Find the values of length at 55°C using quadric interpolation. <table border="1" data-bbox="406 1836 1021 1957"> <tr> <td>T</td><td>40</td><td>50</td><td>60</td></tr> <tr> <td>l</td><td>600.5</td><td>600.6</td><td>600.8</td></tr> </table>	T	40	50	60	l	600.5	600.6	600.8	Application/CO3	6
T	40	50	60								
l	600.5	600.6	600.8								

C)	Develop the algorithm of the Bisection method and simple Trapezoidal numerical Techniques.	Understanding/CO 4	6
	*** End ***		

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